

4.1 Topic project: Applying NLP for topic modelling in a real-life context

NAME: ISHWARYA THANIGAIVELAN

DATE: 16/02/2026

TABLE OF CONTENTS

1. Introduction
 2. Problem Approach
 3. Data preparation and preprocessing
 4. Exploratory Data Analysis
 5. Methods and Techniques
 6. Results and Discussion
 7. Conclusion
- Appendix A - Topic Modelling Visualizations

INTRODUCTION

Topic modeling uncovers hidden themes in text. Techniques like LDA (Latent Dirichlet Allocation), BERTopic, and PHI 4 preprocess and transform text into meaningful topics. LDA uses probabilistic models, BERTopic leverages embeddings and clustering, and PHI 4 enhances topic coherence, enabling insights from large textual datasets efficiently.

PROBLEM APPROACH

This project explores customer reviews of large gym company from Google and Trustpilot to understand recurring issues affecting gym users. Natural language processing (NLP) methods and Machine learning techniques are applied to identify key themes and areas where service improvements can be made across locations.

DATA PREPARATION AND PREPROCESSING

Data was loaded into a dataframe, and rows with missing values in the comment column were removed. Unique and common locations were analyzed, and only English reviews were retained. Google reviews had a shape of (11879, 9) and Trustpilot (15813, 17). Text preprocessing included removing stopwords using NLTK, eliminating non-alphabetic characters and numerals, and tokenizing words with `word_tokenize` for further analysis.

EXPLORATORY DATA ANALYSIS

Frequency distributions were created using `nlk.FreqDist`, with histograms and word clouds generated for the top 10 words in each dataset. Negative reviews (ratings <3) were filtered, and high-frequency non functional words were removed for clarity in visualization but retained for emotion analysis. Across Google and Trustpilot reviews, “Machine” and “Equipment” frequently appear, while positive words like “Good” and “Great” sometimes occur in negative reviews, reflecting mixed sentiments within individual reviews. The frequency distribution of Google review and word cloud for negative review from Trustpilot are shown in Figure 1 and Figure 2 respectively.

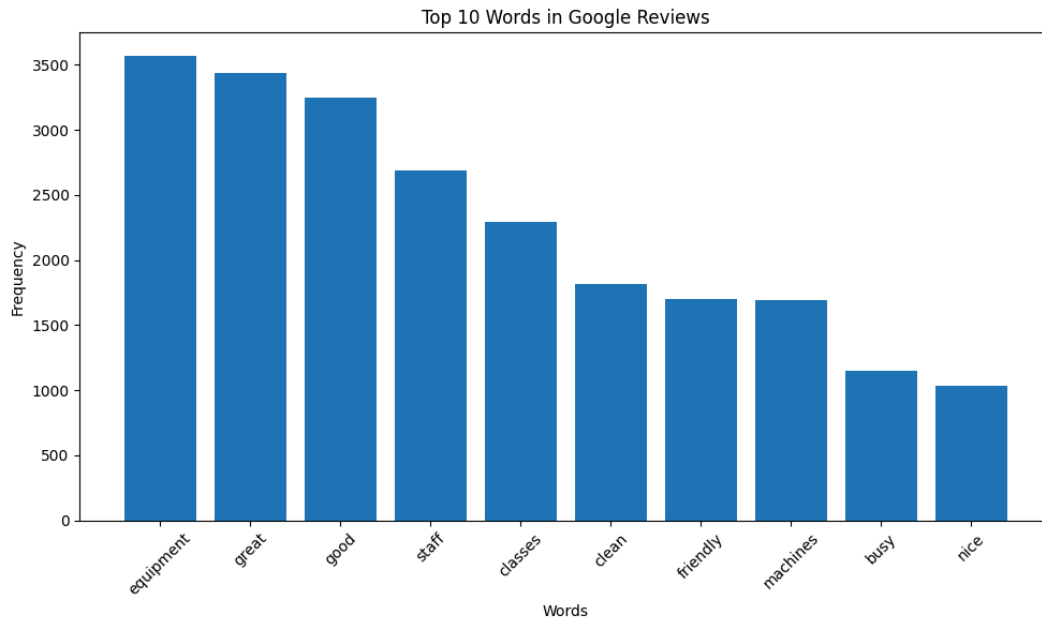


Figure 1:Frequency Distribution of Google Review

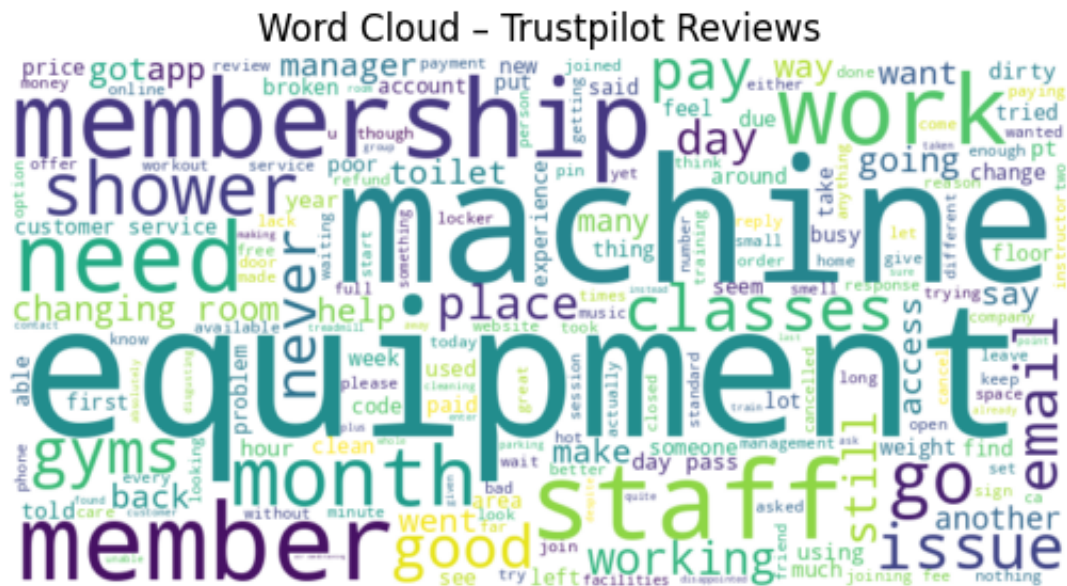


Figure 2: Word cloud for Negative reviews from Trustpilot

METHODS AND TECHNIQUES

Topic modeling was performed using BERTopic on the merged negative reviews dataset. Intertopic distance maps, top-word bar charts, and heatmaps were generated to visualize topic structure and relationships. See Figure A1, Figure A2 and Figure A3 respectively in the Appendix A. The top 20 locations with the highest number of negative reviews were identified. As common locations across platforms were not closely aligned, reviews were aggregated by location and club name, and BERTopic was applied again to the top 30 locations dataset, producing similar visualizations. See Figure A4 for Intertopic distance map and Figure A5 for Heatmap in the Appendix A..

Emotion classification was conducted using BERTopic on negative reviews, revealing anger as the dominant emotion across both datasets. Emotion distribution bar charts were created see Figure A11. A. BERTopic was then applied specifically to anger-related reviews, generating intertopic distance maps, top-word distributions, and heatmaps to analyze key drivers of dissatisfaction. Topic modelling Bar charts, intertopic distance map and heatmap for anger reviews were shown in Figure A6, Figure A7 and Figure A8 respectively in the Appendix A.

Additionally, the large language model Phi-4 Mini Instruct was used to extract the top three topics per review through prompt-based topic extraction. The extracted topic lists were clustered using BERTopic to visualize topic relationships and prominent keywords. See Figure A9 for intertopic distance map in the Appendix A. The same topic lists were further analyzed using Phi-4 Mini Instruct with an actionable prompt to derive concise, practical insights(see the Table T1 in Appendix). Finally, Latent Dirichlet Allocation (LDA) was implemented with standard preprocessing and visualization (See Figure A10 in the Appendix A) to compare probabilistic topic modeling results with BERTopic outputs.

RESULTS AND DISCUSSION

The comparison of topic modeling outputs across multiple methods (custom topic model, LDA, and BERTopic with Phi-4 embeddings) and data subsets (top 30 negative locations and anger-labelled reviews) reveals strong thematic convergence. Across all approaches, the same core drivers of dissatisfaction consistently emerge: cleanliness and hygiene (dirty toilets and changing rooms), staff behaviour and customer service, access issues (PIN codes and entry failures after payment), membership and billing problems, temperature and air conditioning, equipment condition, and operational

availability such as opening hours and class cancellations. This consistency indicates that negative gym experiences are driven by a small number of persistent and systemic service failures rather than isolated incidents.

The word cloud for the merged and sorted top-30 locations run appears broadly similar to the initial negative-review run, highlighting frequent terms such as equipment, machine, staff, member, and membership. However, additional emphasis is placed on showers, people, and toilets, suggesting that hygiene and shared facility usage issues are more salient in the worst-performing locations. This supports the clustering results, which show that poorly rated sites are characterised by tangible infrastructure and cleanliness failures.

Differences between models mainly concern the level of granularity. The custom topic model provides more operationally actionable distinctions, such as separating toilet cleanliness from ventilation and smell, and class booking problems from general access issues. In contrast, the Phi-4 BERTopic output aggregates multiple operational issues into broader experiential themes, which is useful for high-level monitoring but less precise for targeted interventions. LDA and the top-30 location clusters place stronger emphasis on physical infrastructure problems, including broken equipment, showers, temperature control, and overcrowding.

Synthesising findings across models supports actionable recommendations: improving cleanliness and hygiene standards; reducing overcrowding during peak hours; providing customer service training; improving ventilation and air-conditioning; maintaining equipment and discouraging misuse during busy periods; and making membership cancellation clearer and more customer-friendly. Anger-focused topics further indicate that perceived unfairness and unreliable service are key triggers of strong negative emotions and churn risk.

CONCLUSION

Across the workflow—negative-review BERTopic, anger-emotion BERTopic, Phi-4 BERTopic, Phi-4 actionable insights, and LDA—the same major problems consistently emerge: poor cleanliness and hygiene, poorly maintained equipment and facilities (including showers and temperature control), access and membership issues, and negative staff behaviour. These issues appear across general and anger-focused reviews, indicating that customer dissatisfaction is driven by systemic service failures rather than isolated incidents.

The different models complement each other effectively. Negative-review and anger-emotion BERTopic highlight operational and emotional triggers, LDA structures complaints into broader themes, and Phi-4 embeddings provide high-level, actionable

insights. Together, they offer a comprehensive understanding of both the practical and emotional aspects of customer dissatisfaction. The Phi-4 actionable insights further translate these findings into concrete recommendations, including improving hygiene standards, maintaining equipment, enhancing ventilation and air-conditioning, training staff in customer service, managing overcrowding, and clarifying membership cancellation procedures. This integrated multi-model approach provides a robust framework for diagnosing service failures, reducing customer frustration, and prioritising targeted improvements across the gym network.

Appendix A - Topic Modelling Visualizations

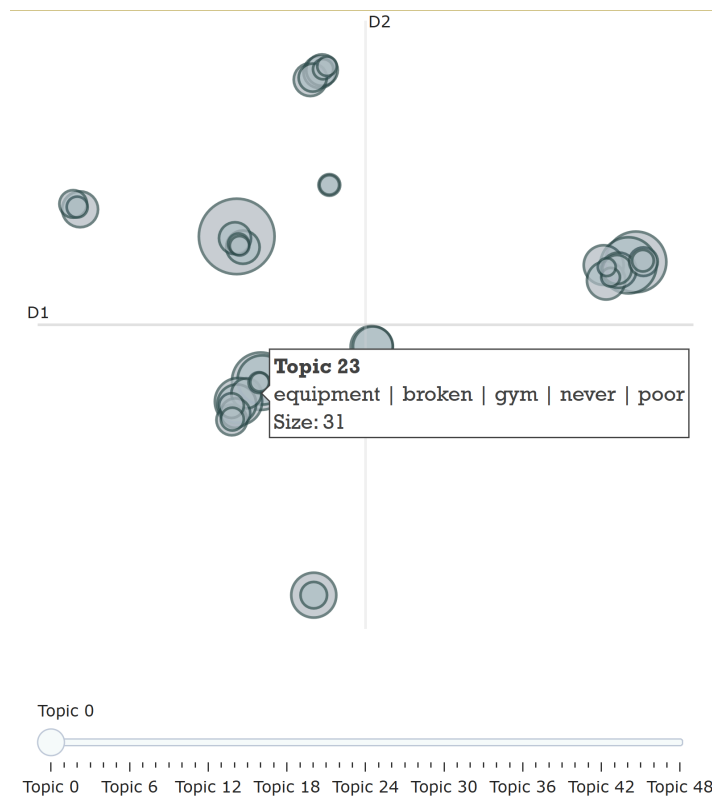


Figure A1: Intertopic distance Map for Merged Negative Reviews

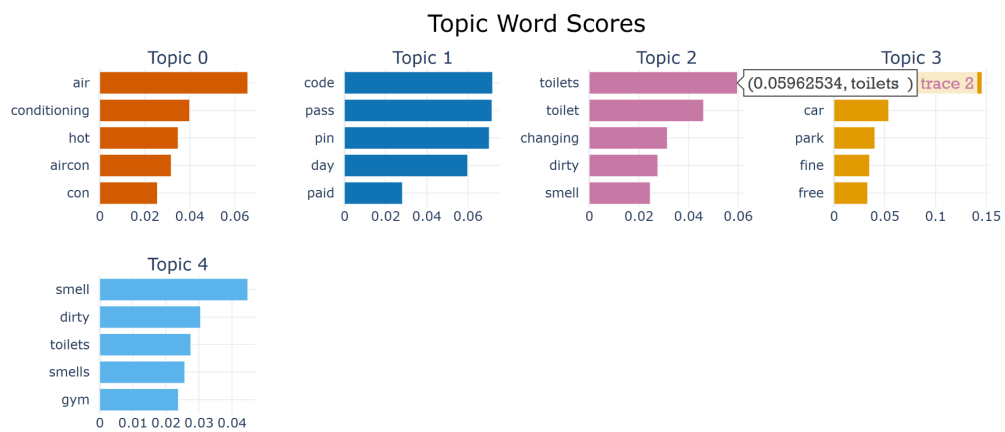


Figure A2: Barchart for Topwords in Topics for Merged Negative Reviews

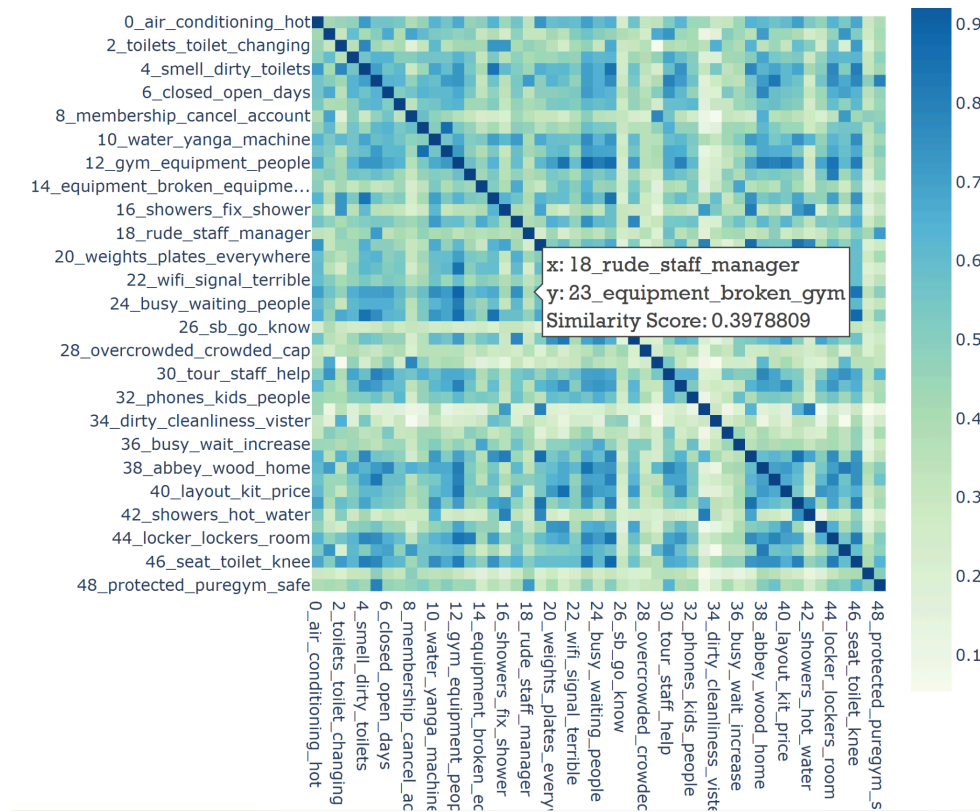


Figure A3: Heatmap for Merged Negative Reviews

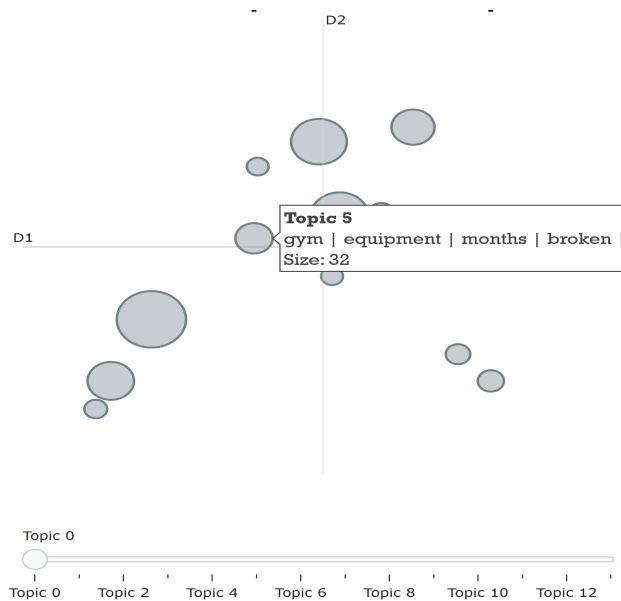


Figure A4: Intertopic Distance Map for Top30 Location Topic Modelling

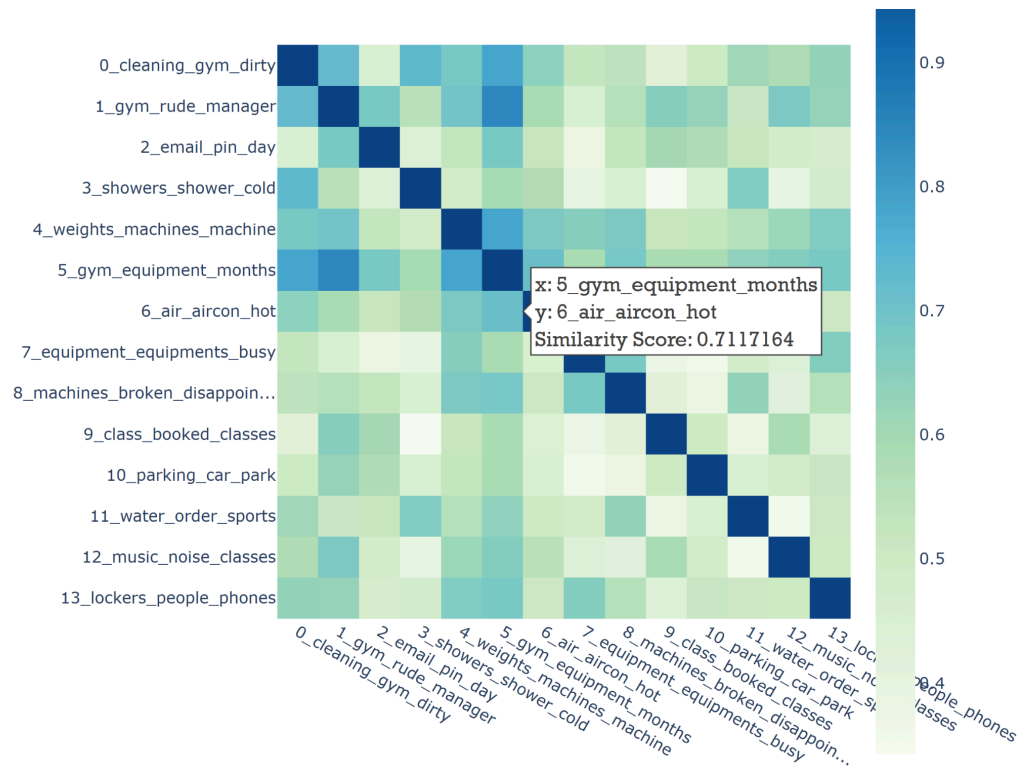


Figure A5: Heatmap for Top30 Location Topic Modelling



Figure A6 : Anger Review Topwords in Topics

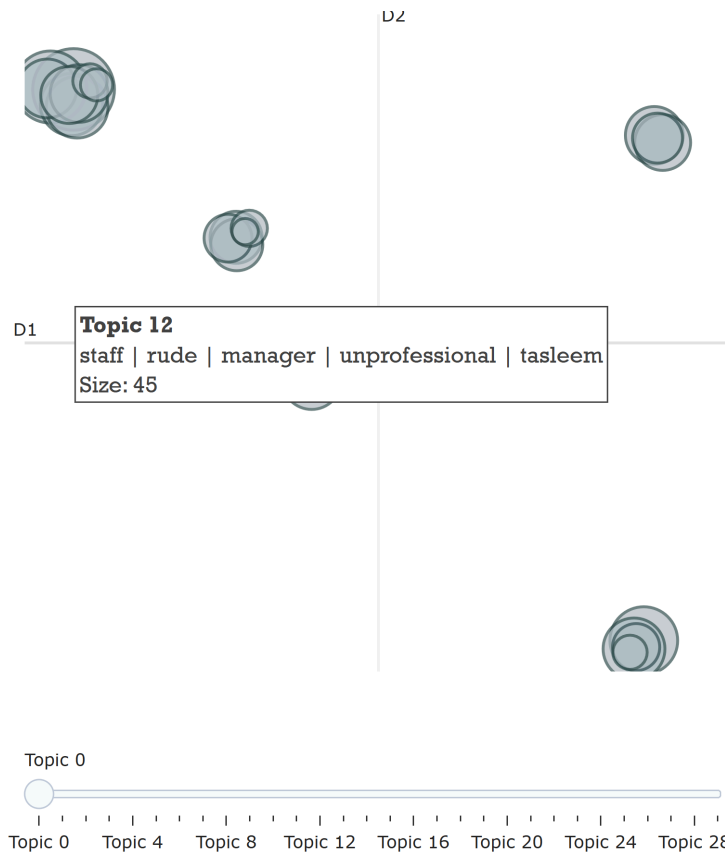


Figure A7: Intertopic Distance Map for Anger Reviews

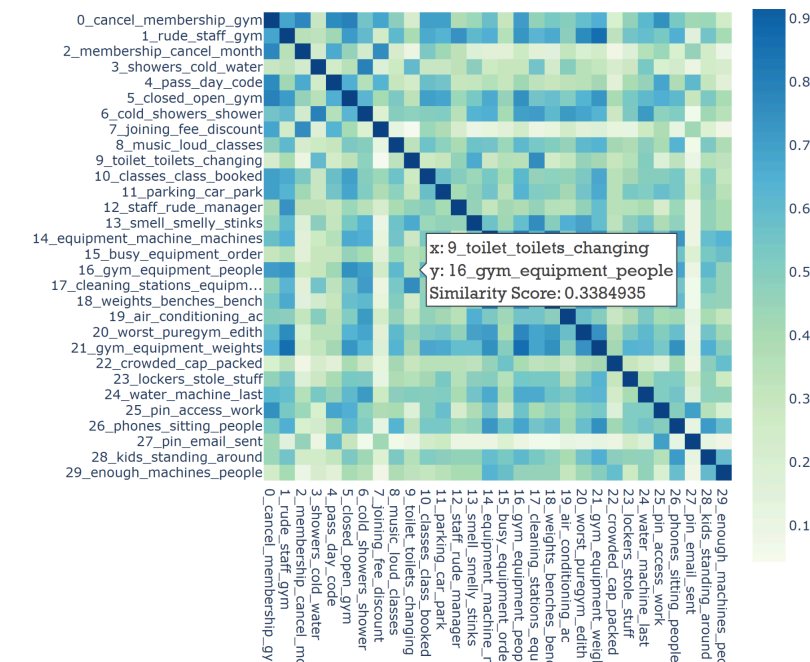


Figure A8: Heatmap for Anger Reviews

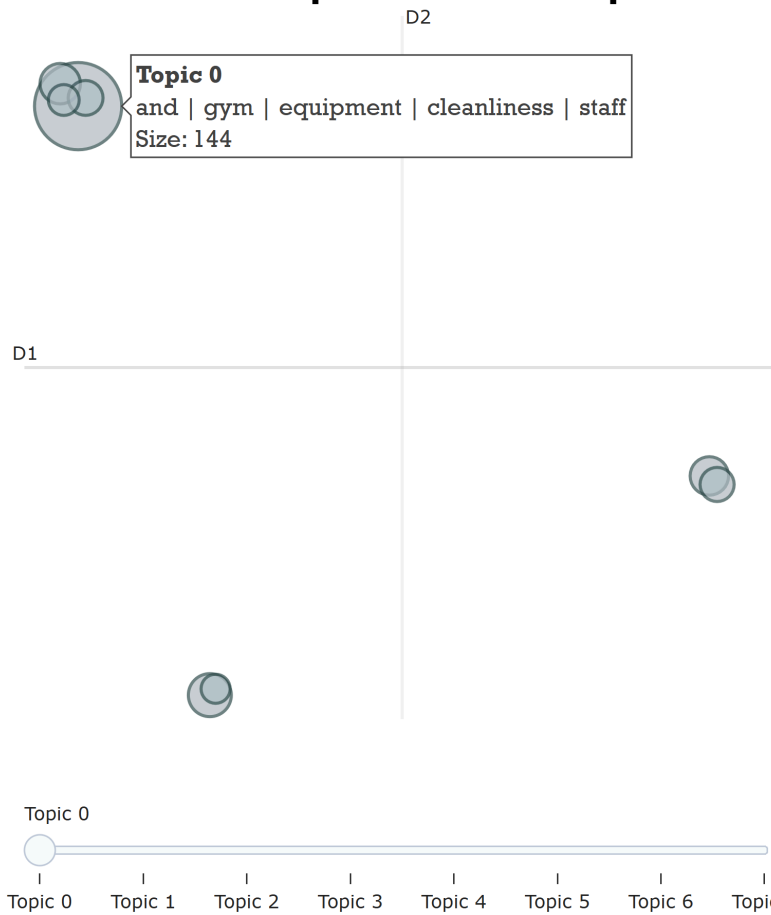


Figure A9: Intertopic Distance Map for Phi 4 Output

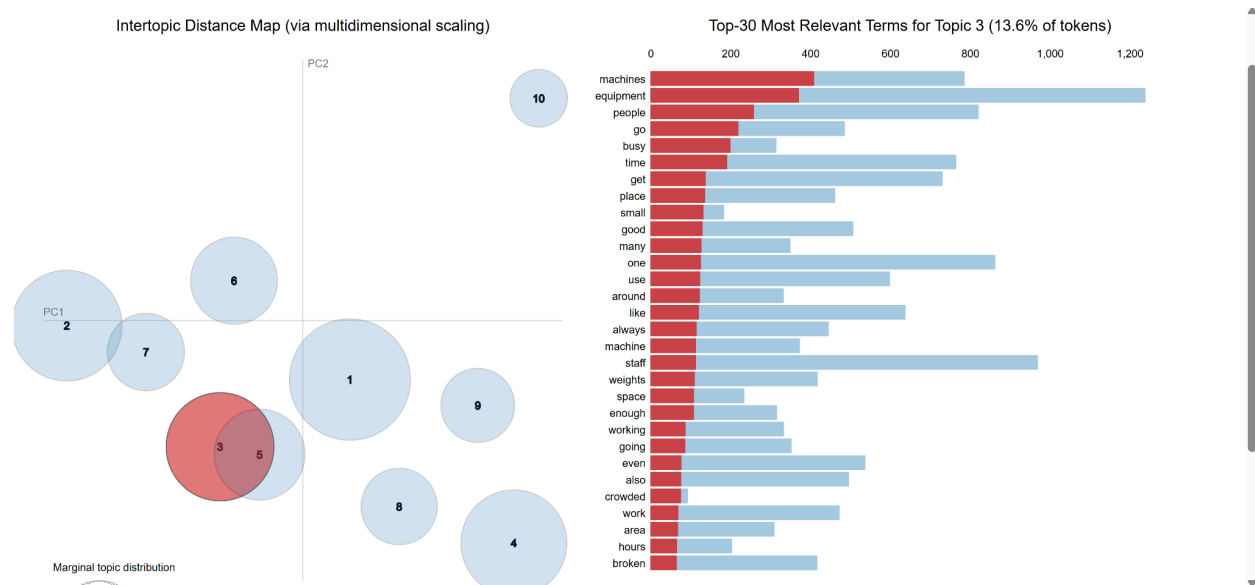


Figure A10: LDA Visualisation

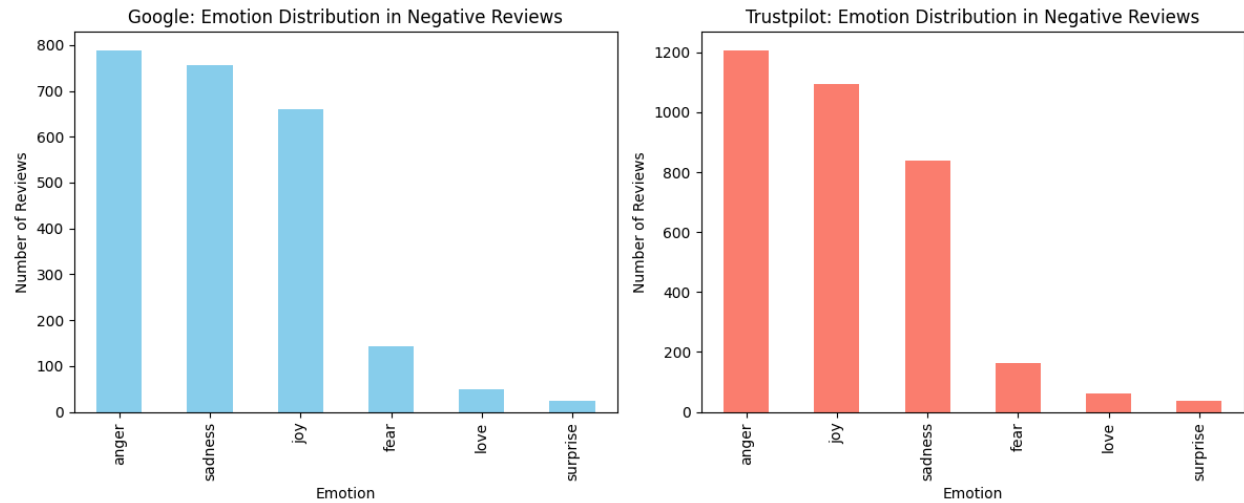


Figure A11: Emotion Distribution Barchart

Table T1

Final Actionable Insights for the Gym Company:

1. Improve cleanliness and hygiene standards across the gym, including changing rooms and bathrooms.
2. Reduce overcrowding during peak hours by limiting group usage and improving crowd management.
3. Provide customer service training to staff to improve friendliness and professionalism.
4. Improve ventilation and air-conditioning to ensure a comfortable workout environment.
5. Ensure gym equipment is well-maintained and reduce misuse of machines during busy periods.
6. Make the membership cancellation process clearer and more customer-friendly.

